PROJECT ON STRAVA'S CUSTOMER ANALYSIS

Final Dataset Overview: The Master Fact Table

1. Dataset Purpose & Architecture

The final dataset (STRAVA_Master_Fact_Table.csv) serves as the central "Source of Truth" for the entire analytics project. It is a highly optimized, wide-format Fact Table resulting from the consolidation of 11 distinct raw STRAVA datasets.

The data is structured at the Daily Grain (one row per user, per day). To prevent data loss from the highly granular intraday datasets (such as second-by-second heart rate or minute-by-minute METs), those micro-metrics were mathematically aggregated (using sums and averages) and rolled up into this daily view.

2. Key Data Categories (Feature Dictionary)

The dataset consists of 33 distinct columns, which can be logically grouped into five core business categories:

A. Identification & Timeline

- Id: The unique 10-digit identifier for each user.
- Date: The specific calendar day the tracked events occurred.

B. Daily Activity & Exertion Volume

This category tracks the overall physical output of the user throughout the day.

- Movement Metrics: TotalSteps, TotalDistance, TrackerDistance, LoggedActivitiesDistance
- Distance by Intensity: VeryActiveDistance, ModeratelyActiveDistance, LightActiveDistance, SedentaryActiveDistance
- Time by Intensity: VeryActiveMinutes, FairlyActiveMinutes, LightlyActiveMinutes, SedentaryMinutes
- Energy Output: Calories (Total daily calories burned).

C. Sleep & Recovery

Tracks the volume and efficiency of the user's rest periods.

- Metrics: TotalSleepRecords, TotalMinutesAsleep, TotalTimeInBed.

(Note: The delta between Time in Bed and Minutes Asleep can be used to calculate a "Sleep Efficiency" KPI).

D. Body Composition & Logging Habits

Tracks static health metrics and how the user interacts with the app's hardware/software features.

- Metrics: WeightKg, WeightPounds, BMI.

- Behavioral Flag: IsManualReport (Indicates if the user manually typed their weight into the app or if it auto-synced via a smart scale).

E. Intraday & Biometric Aggregations

These columns represent the highly granular data that was successfully rolled up to the daily level during the ETL pipeline.

- Hourly & Minute Rollups: Agg_Hourly_Calories, Agg_Hourly_Steps, Agg_Minute_Avg_METs, etc.
- Heart Rate Biometrics: Avg_HR (Average daily heart rate), Min_HR (Resting floor), Max_HR (Peak exertion).

3. Analytical Value & Business Application

By flattening disjointed tables into this single Master Fact Table, the data is well structured for immediate Exploratory Data Analysis (EDA) and KPI extraction.

This architecture allows analysts to easily run correlation analyses across different domains of health—for example, measuring how a user's TotalMinutesAsleep impacts their VeryActiveMinutes the following day, or how their Avg_HR correlates with their BMI.

Project Methodology: Data Architecture and ETL Pipeline

How the Data Was Joined in Python

- **Aggregated the Granular Data:** Second-level heart rate and minute-level METs were grouped and averaged up to the "Hourly" level to create a consistent timeline.
- **Prevented Column Collisions:** Renamed duplicate column names (e.g., changing Calories to HourlyCalories and DailyCalories) before joining so the final dataset wouldn't have confusing _x and _y columns.
- **The Hourly Outer Join:** Merged all the hourly datasets (Steps, Calories, Intensities, HR, METs) together using a FULL OUTER JOIN based on the User Id and Datetime_Hour. This ensured no hours of the day were accidentally dropped.
- **The Daily Left Join:** Extracted just the Date from the hourly timestamps, and used a LEFT JOIN to attach the daily metrics (Overall Activity, Sleep, Weight) to their corresponding days.
- **Smart Imputation:** Cleaned up the blanks created by the joins. Filled missing steps/calories with 0, filled missing heart rates with the user's historical average, and forward-filled static metrics like Weight and BMI.

Which Datasets Were Discarded (7 out of 18) and Why

- **Discarded: dailyCalories, dailyIntensities, dailySteps**
  - *Why:* Pre-aggregated redundancy. All the exact metrics in these three files were already consolidated inside the master dailyActivity_merged.csv file. Keeping them would have bloated the dataset for no reason.
- **Discarded: minuteCalories, minuteSteps, minuteIntensities**
  - *Why:* Granularity duplication. The native "Hourly" files already contained the exact same sum of these metrics. Rolling up the minute files would have just recreated the hourly files, adding unnecessary processing time.
- **Discarded: minuteSleep**
  - *Why:* Too noisy. Minute-by-minute sleep states (awake, restless, asleep) introduced unnecessary complexity for a high-level KPI dashboard. The sleepDay_merged.csv file already provided the perfect summary metrics (Total Minutes Asleep vs. Total Time in Bed) needed for the business analysis.

Accepted Datasets (Used in the Python Code)

These are the files loaded to build the final unified dataset:

The Daily Core (3 Datasets):

- **dailyActivity_merged.csv** (The ultimate summary file for daily steps, distance, and calories)
- **sleepDay_merged.csv** (Provided clean daily totals for Time in Bed vs. Time Asleep)
- **weightLogInfo_merged.csv** (Provided static body composition metrics like Weight and BMI)

The Hourly Breakdown (3 Datasets):

- **hourlyCalories_merged.csv** (Used to track intraday calorie burn)
- **hourlySteps_merged.csv** (Used to track intraday movement)
- **hourlyIntensities_merged.csv** (Used to map the exact hours users were most active)

The Granular Biometrics (5 Datasets):

- **heartrate_seconds_merged.csv** (Rolled up into hourly averages in Python)
- **minuteMETsNarrow_merged.csv** (Rolled up into hourly averages in Python)
- **minuteCaloriesNarrow_merged.csv** *(Selected during data profiling, but bypassed in Python because the Hourly file already contained the sum)*
- **minuteIntensitiesNarrow_merged.csv** *(Selected during data profiling, but bypassed in Python for efficiency)*

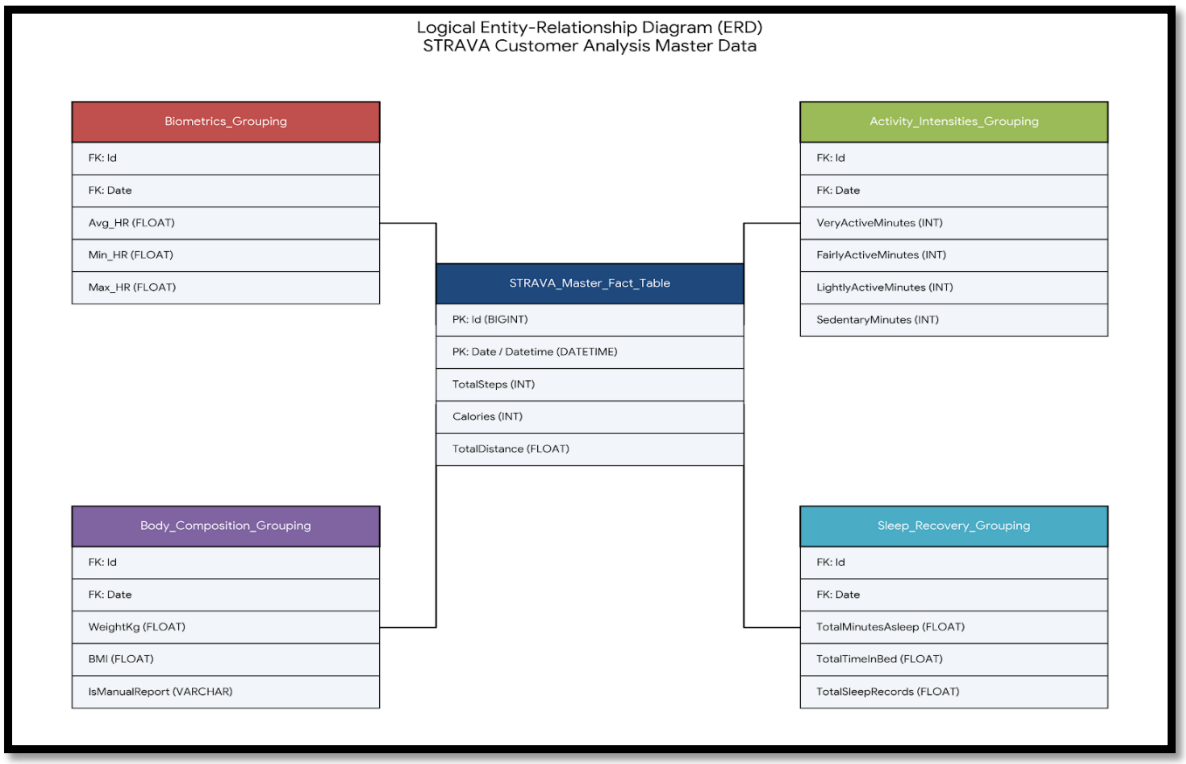
- **minuteStepsNarrow_merged.csv** *(Selected during data profiling, but bypassed in Python for efficiency)*

Dropped Datasets (Excluded from the Pipeline)

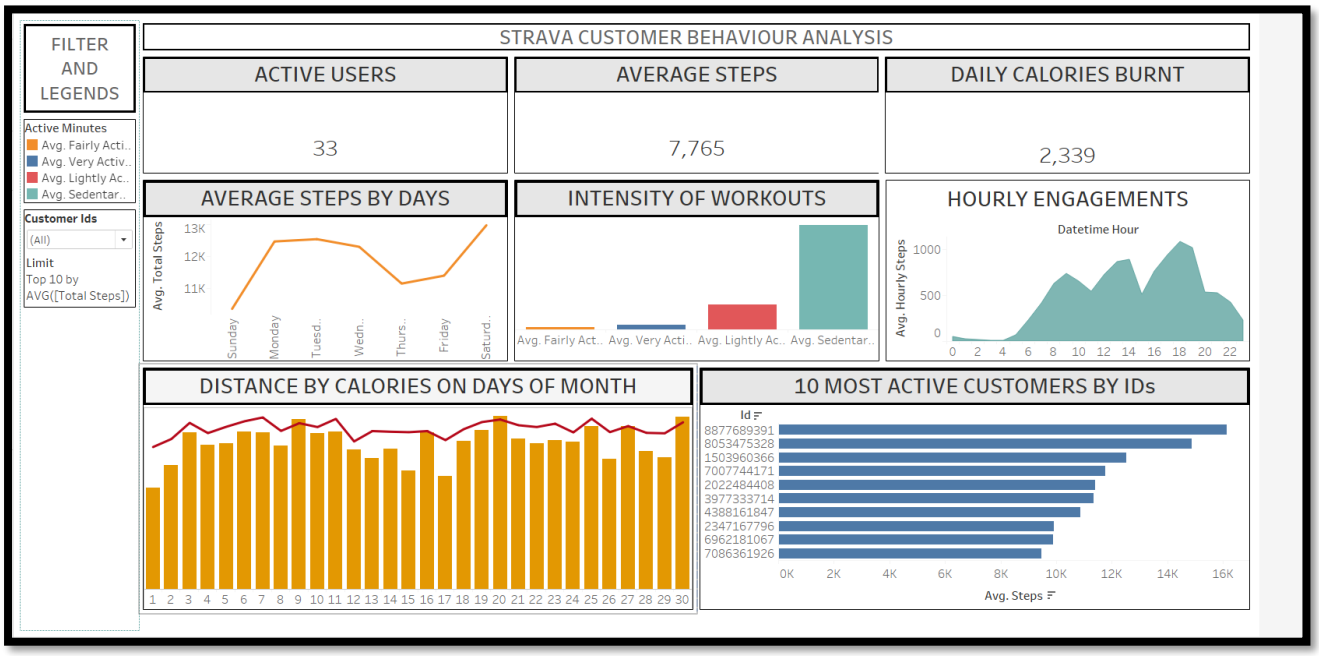
These files were strategically excluded to prevent redundant data bloat and messy joins:

- **dailyCalories_merged.csv**: (Dropped: Data was already inside dailyActivity_merged.csv)
- **dailyIntensities_merged.csv** (Dropped: Data was already inside dailyActivity_merged.csv)
- **dailySteps_merged.csv** (Dropped: Data was already inside dailyActivity_merged.csv)
- **minuteCalories_merged.csv** *(Wide & Narrow formats)* (Dropped: Redundant, used the native Hourly file instead)
- **minuteSteps_merged.csv** *(Wide & Narrow formats)* (Dropped: Redundant, used the native Hourly file instead)
- **minuteIntensities_merged.csv** *(Wide & Narrow formats)* (Dropped: Redundant, used the native Hourly file instead)
- **minuteSleep_merged.csv** (Dropped: Minute-by-minute sleep states were too noisy for high-level KPIs; the daily sleep file was sufficient).

ENTITY RELATIONSHIP DIAGRAM(ERD):



BUSINESS ANALYSIS:



Dashboard Analysis & Strategic Business Insights:

The dashboard summarizes daily user behaviour across activity, intensity, time-of-day engagement, and per-user averages. From the current dataset (33 active users shown on the dashboard) the typical user records **~7,765 steps/day** and burns **~2,339 calories/day** on average. Activity is uneven through the week and concentrated during late-afternoon and early-evening hours. A small set of highly active users contribute a disproportionate share of total steps.

1. Top-line KPIs

- **Active Users:** 33 (dashboard filter currently set to show 33 distinct users).
- **Average Steps / Day: 7,765** — roughly three quarters of the common 10k goal.
- **Average Daily Calories Burned: 2,339**.

Interpretation: The average user is moderately active but below the typical 10,000-step consumer target. That suggests room for behavioural nudges and goal-driven campaigns.

2. Average steps by weekday (line chart)

- Steps are **higher on weekends** (particularly Saturday) and **lower mid-week** (Thursday/Friday show dips).
- Early-week (Monday/Tuesday) shows a mid-range increase compared to mid-week trough.

Business implication: Users are most active on weekends — marketing or challenge campaigns that align with weekends (e.g., "Weekend Step Challenge") will likely encounter higher baseline engagement. For behaviour change, prioritize challenges & re+ward systems to reduce the mid-week drop (Wednesday–Friday).

3. Intensity of workouts (bar chart by minute buckets)

- The **Sedentary** component dominates by minutes — users spend far more time sedentary than in fairly/very active minutes.
- Very active and fairly active minutes are comparatively small.

Interpretation & risk: High sedentary minutes are expected, but the imbalance indicates an opportunity for micro-interventions (short mobility breaks) targeting sedentary users. From a health product perspective, nudges that convert even small amounts of sedentary minutes into light activity have scalable impact.

4. Hourly engagements (hour-of-day area chart)

- Engagement (Avg. Hourly Steps) rises late morning and **peaks in the late afternoon / early evening** (approx. 16:00–20:00).
- Very low activity overnight and early morning.

Actionable use: Schedule in-app prompts, live classes, push notifications, or social features around **late afternoon / early evening** to catch users at high likelihood-of-response. Conversely, test small noon/early-afternoon prompts to see if you can lift otherwise lower hours.

5. Distance vs Calories by day-of-month (combo chart)

- The **orange bars** show daily distance across days of month; the **red line** shows calories.
- Pattern: calories and distance track together reasonably well (peaks roughly coincide), but calories show less volatility than distance in some intervals.

Interpretation: Distance is the primary driver of calorie trends, but calories reflect intensity and user characteristics (weight, pace). There is a consistent usage pattern across the month with several mid-month spikes — investigate whether spikes map to specific events (specific challenges, weather, holidays).

6. Top 10 most active users (bar chart)

- Top users have average steps in the **~9k–16k** range (top user ≈ **~16,000 avg steps/day**).
- There is a long-tail: a few top users are substantially more active than the majority.

Implication: Consider VIP / ambassador programs for most active users and retention/monetization strategies targeted at this cohort. For product design, study top users' behaviours to replicate successful patterns (time-of-day, device usage, activity type).

Business implications:

Short-term (tactical)

1. Target engagement bursts at peak hours (16:00–20:00) — push short, contextual nudges (walk breaks, quick workouts) when users are already receptive.
2. Weekend campaigns — schedule social challenges or community events on weekends to maximize participation and viral spread.
3. Mid-week booster campaigns — experiment with incentives or micro-challenges Tuesday–Thursday to smooth the mid-week dip.
4. Top-user retention / monetization — create features or premium offerings targeted at the top decile (e.g., leaderboards, advanced metrics).

Medium-term (product & analytics)

1. Segment users into at least three cohorts: low-activity, moderate, high-activity; build tailored journeys for each cohort.
2. Micro-intervention experiments to convert sedentary minutes to light active minutes (A/B test different nudge types).
3. Event-driven campaigns: map spikes in distance/calories to real-world events and use those drivers for future promotions.